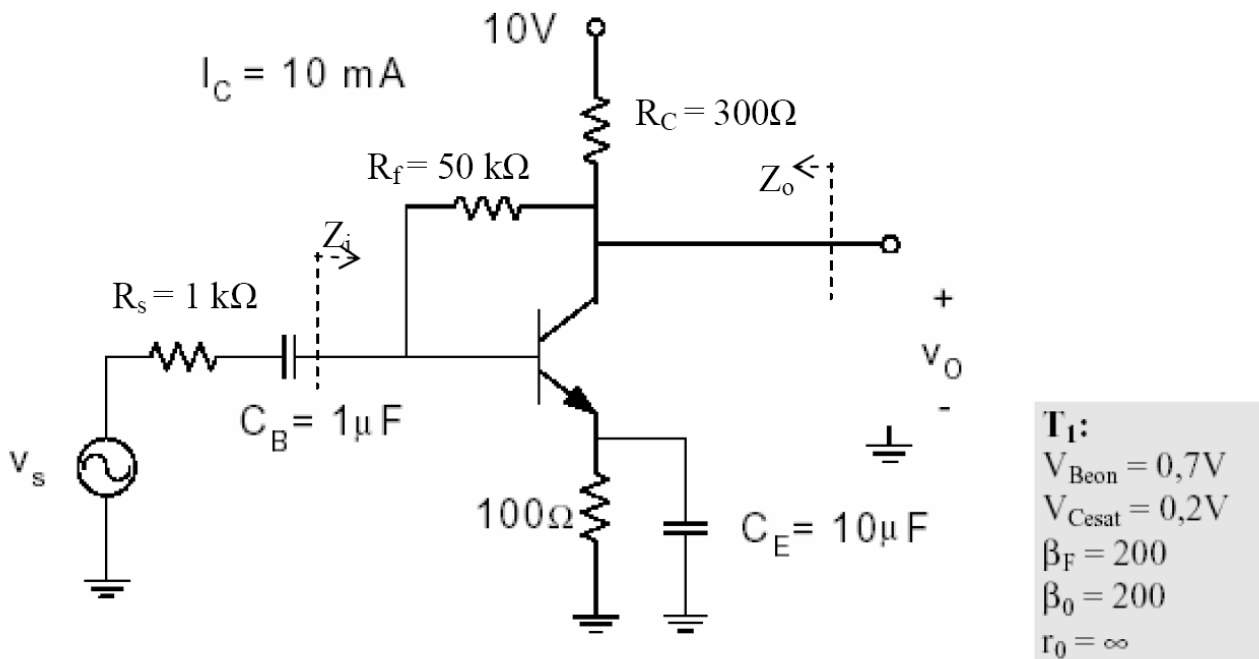


RE1 – Feedback amplifier with bipolar transistor



The circuit of figure represents a feedback amplifier.

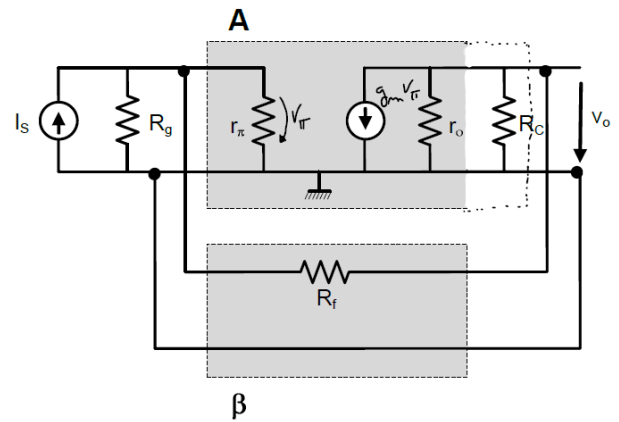
- Identify the feedback topology and represent the corresponding models of blocks A and β .
- Calculate the gain of the loaded amplifier A' and the gain β' of the feedback loop, in the intermediate frequency range.
- Based on feedback theory, determine the expression for V_o/V_s .
- Calculate the input impedance Z_i and the output impedance Z_o of the feedback amplifier.
- Sketch the frequency responses of the gain modulus of the basic amplifier and the feedback amplifier. Assume that the lower and upper cut-off frequencies of the loaded basic amplifier have the following values: $f_L = 500\text{Hz}$ and $f_H = 400\text{kHz}$.

Resolution

a)

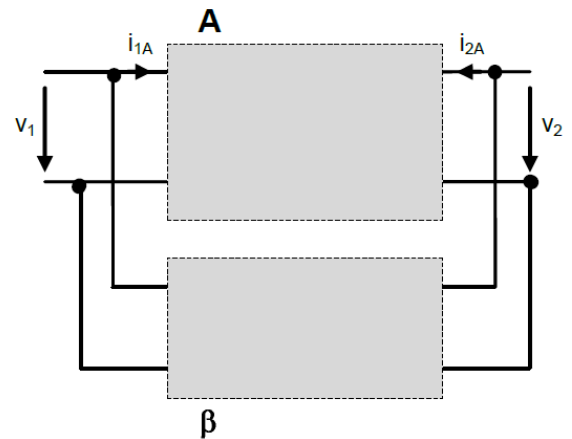
Two-port A senses the output voltage and makes a difference between input currents:

Parallel → Parallel
or Current → Voltage
or Transimpedance amplifier



b)

Common input and output variables: voltages v_1 and v_2 .
They must therefore be independent variables.
The representation should be made using the
Admittances matrix:



$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = [Y] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Matrix A

$$y_{11A} = \frac{i_1}{v_1} \Big|_{v_2=0} = \frac{1}{r_{\pi}}$$

$$y_{12A} = \frac{i_1}{v_2} \Big|_{v_1=0} = 0$$

$$y_{21A} = \frac{i_2}{v_1} \Big|_{v_2=0} = g_m$$

$$y_{22A} = \frac{i_2}{v_2} \Big|_{v_1=0} = \frac{1}{r_o // R_C}$$

Matrix β

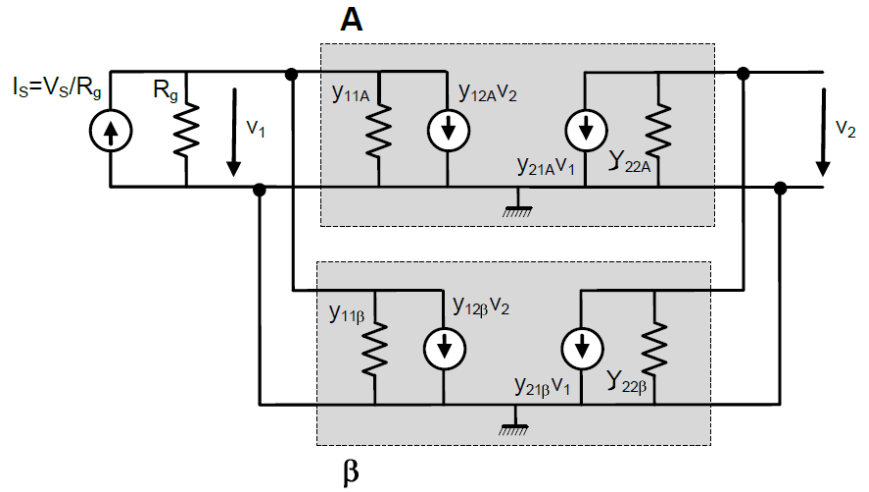
$$y_{11\beta} = \frac{1}{R_f}$$

$$y_{12\beta} = -\frac{1}{R_f}$$

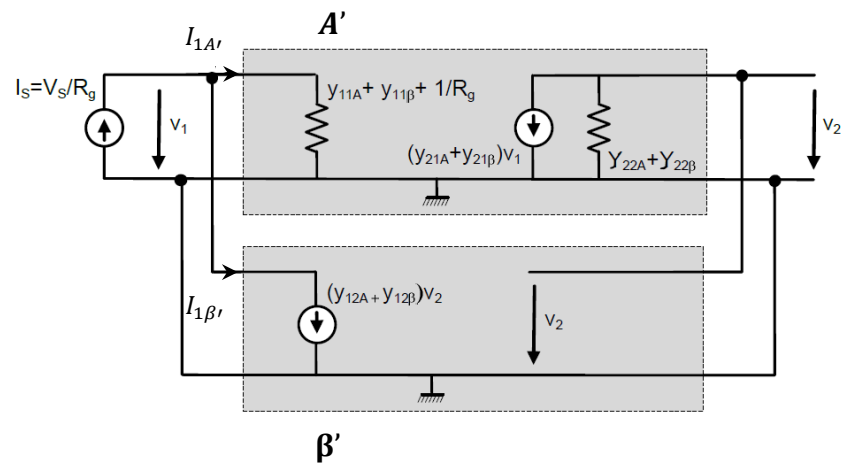
$$y_{21\beta} = -\frac{1}{R_f}$$

$$y_{22\beta} = \frac{1}{R_f}$$

Norton equivalent of V_S, R_g :



Due to the representation chosen for A and β (with common v_1 and v_2) it is easy to load A and obtain both one-way two-access networks:



Gain A' – Transimpedance (open loop):

$$A' = \frac{V_2}{I_{1A'}} = -\frac{y_{21A} + y_{21\beta}}{\left(y_{11A} + y_{11\beta} + \frac{1}{R_g}\right)(y_{22A} + y_{22\beta})} = -\frac{g_m - \frac{1}{R_f}}{\left(\frac{1}{r_\pi} + \frac{1}{R_f} + \frac{1}{R_g}\right)\left(\frac{1}{r_o} + \frac{1}{R_C} + \frac{1}{R_f}\right)} \approx -\frac{g_m}{\left(\frac{1}{r_\pi} + \frac{1}{R_g}\right)\left(\frac{1}{R_C}\right)}$$

$$A' = 44.7k\Omega$$

$$\left(I_C + \frac{I_C}{\beta}\right)R_C + \frac{I_C}{\beta}R_f + R_E \frac{\beta + 1}{\beta}I_C + V_{BEon} = V_{CC}$$

$$V_{BEon} = 0.7V \Rightarrow I_C = 14.26mA$$

$$g_m = \frac{I_C}{V_T} = \frac{14.26mA}{25mV} = 584mS$$

$$R_\pi = \frac{\beta_o}{g_m} = 542.5\Omega$$

This expression is similar for all topologies - all we need to do is determine the y_A and y_β matrices.

Feedback β' – transconductance:

$$\beta' = \frac{I_{1\beta'}}{V_2} = y_{12A} + y_{12\beta} = -\frac{1}{R_f} = -20\mu S$$

c)

Total gain including feedback (A_f):

$$A_f = \frac{V_2}{I_S} = \frac{A'}{1 + A'\beta'} = -23.6k\Omega$$

Voltage gain:

$$K_v = \frac{V_2}{V_S} = \frac{V_2}{R_g I_S} = \frac{A_f}{R_g} = -23.6$$

d)

Open loop input impedance ($Z_{iA'}$):

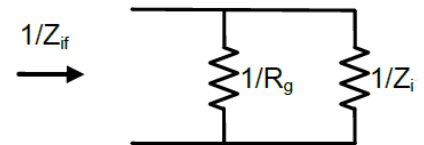
$$Z_{iA'} = \frac{1}{y_{11A} + y_{11\beta} + \frac{1}{R_g}} = \frac{1}{\frac{1}{r_\pi} + \frac{1}{R_f} + \frac{1}{R_g}} = 253.8\Omega$$

Input impedance with feedback (Z_{if}):

$$Z_{if} = \frac{Z_{iA'}}{1 + A'\beta'} = 134\Omega$$

To calculate the actual input impedance, the effect of R_g must be discounted (input in parallel the discount has to be made with admittances):

$$Z_i = \frac{1}{\frac{1}{Z_{if}} - \frac{1}{R_g}} = 154.8\Omega$$



$$\frac{1}{Z_{if}} = \frac{1}{R_g} + \frac{1}{Z_i}$$

Open loop output impedance ($Z_{oA'}$):

$$Z_{oA'} = \frac{1}{y_{22A} + y_{22\beta} + \frac{1}{R_L}} = \frac{1}{\frac{1}{r_o} + \frac{1}{R_c} + \frac{1}{R_f}} = 298.2\Omega$$

Output impedance with feedback (Z_{of}), as there is no load resistor ($R_L = \infty$), $Z_o = Z_{of}$:

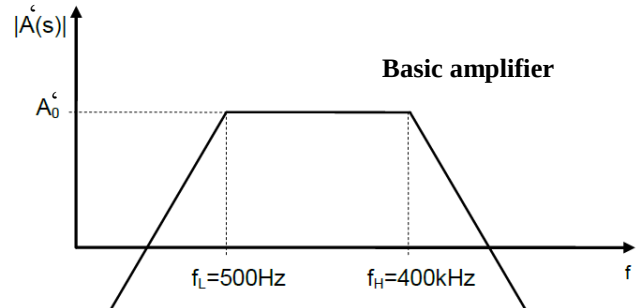
$$Z_o = \frac{Z_{oA'}}{1 + A'\beta'} = 157.5\Omega$$

1. Feedback and stability

e)

Basic amplifier

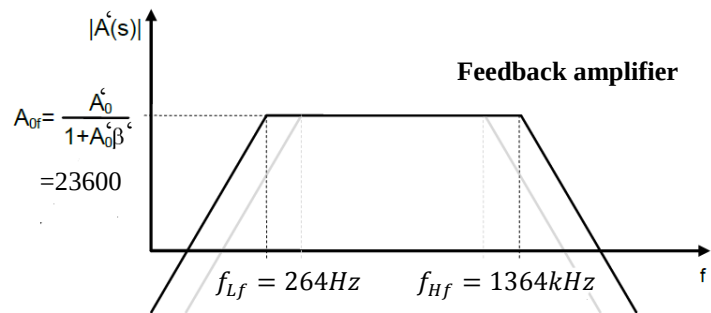
$$A(s) = \frac{A_0' \frac{s}{\omega_L}}{\left(1 + \frac{s}{\omega_L}\right) \left(1 + \frac{s}{\omega_H}\right)}$$



Feedback amplifier

$$A_f(s) = \frac{A(s)}{1 + A(s)\beta}$$

$$A_f(s) = \frac{\frac{A_0' \frac{s}{\omega_L}}{\left(1 + \frac{s}{\omega_L}\right) \left(1 + \frac{s}{\omega_H}\right)}}{1 + \frac{A_0' \frac{s}{\omega_L}}{\left(1 + \frac{s}{\omega_L}\right) \left(1 + \frac{s}{\omega_H}\right)} \beta} = 23600$$



$$A_f(s) = \frac{A_0' \frac{s}{\omega_L}}{\left(1 + \frac{s}{\omega_L}\right) \left(1 + \frac{s}{\omega_H}\right) + A_0' \frac{s}{\omega_L} \beta}$$

$$1 + A_0' \beta = 1.894$$

Low frequencies ($\omega \ll \omega_H$):

$$A_f(s) \approx \frac{A_0' \frac{s}{\omega_L}}{1 + \frac{s}{\omega_L / (1 + A_0' \beta)}} = \frac{A_0' \frac{s}{\omega_L}}{1 + \frac{s}{\omega_{Lf}}}$$

$$\boxed{\omega_{Lf} = \frac{\omega_L}{1 + A_0' \beta}} \Rightarrow f_{Lf} = 264\text{Hz}$$

Low frequencies ($\omega \gg \omega_L$):

$$A_f(s) \approx \frac{A_0'}{\left(1 + \frac{s}{\omega_H}\right) + A_0' \beta} = \frac{\frac{A_0'}{1 + A_0' \beta}}{1 + \frac{s}{\omega_H (1 + A_0' \beta)}} = \frac{\frac{A_0'}{1 + A_0' \beta}}{1 + \frac{s}{\omega_{Hf}}}$$

$$\boxed{\omega_{Hf} = \omega_H (1 + A_0' \beta)}$$

$$\Rightarrow f_{Hf} = 1364\text{kHz}$$

Bandwidth without feedback:

$$\frac{\Delta B}{2\pi} = f_H - f_L = 399.5\text{kHz}$$

Bandwidth with feedback:

$$\frac{\Delta B_f}{2\pi} = f_{Hf} - f_{Lf} = 1363.7\text{kHz}$$